CECH COHOMOLOGY OF AN INJECTIVE MODULE IS CONCENTRATED IN DEGREE ZERO

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for all degrees, the positive degrees being reduced to the extension property and the degree zero case read off from the same description.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Cech cohomology of a covering with coefficients in a module. Same chapter, the Cech complex of a covering and its cohomology, and its description by maps from representable objects. Supplies the groups computed in the statement and the identification used at the start of the proof.
- Injectivity of a module over a ringed site and the resulting extension property. Same chapter, the extension property characterising injective modules, applied to maps from subsheaves of representable sheaves. Supplies the vanishing argument for the higher Cech groups.

Source reference. Original lemma 03AS: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 971–971. Navigation pointer invoked at proof line 1833; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03CU: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1850–1850. Navigation pointer invoked at proof line 1850; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 03FB: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1770–1770. Navigation pointer invoked at proof line 1851; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03F5: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1150–1150. Navigation pointer invoked at proof line 1856; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03AN: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 851–851. Navigation pointer invoked at proof line 1858; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{U} = \{U_i \to U\}_{i \in I}$ be a covering of $\mathcal{C}$. Let $\mathcal{I}$ be an injective $\mathcal{O}$-module, i.e., an injective object of $Mod(\mathcal{O})$. Then*

$$\check{H}^p(\mathcal{U}, \mathcal{I}) = \begin{cases} \mathcal{I}(U) & \text{if} \quad p = 0 \\ 0 & \text{if} \quad p > 0 \end{cases}$$

Proof. Lemma 03AS gives the first equality in the following sequence of equalities

$$\begin{aligned} \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{I}) &= \mathrm{Mor}_{PAb(\mathcal{C})}(\mathbf{Z}_{\mathcal{U},\bullet}, \mathcal{I}) \\ &= \mathrm{Mor}_{PMod(\mathbf{Z})}(\mathbf{Z}_{\mathcal{U},\bullet}, \mathcal{I}) \\ &= \mathrm{Mor}_{PMod(\mathcal{O})}(\mathbf{Z}_{\mathcal{U},\bullet} \otimes_{p,\mathbf{Z}} \mathcal{O}, \mathcal{I}) \end{aligned}$$

The third equality by Modules on Sites, Lemma 03CU. By Lemma 03FB we see that $\mathcal{I}$ is an injective object in $PMod(\mathcal{O})$. Hence $\mathrm{Hom}_{PMod(\mathcal{O})}(-, \mathcal{I})$ is an exact functor. By Lemma 03F5 we see the vanishing of higher Čech cohomology groups. For the zeroth see Lemma 03AN. □